ESE 123 Lab 2 Report: KVL and DMM Measurements

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Objective. Check KVL with two power supply channels in series, measure current and resistance with the DMM, see how the test leads, the human body and a powered circuit affect readings, and see the effect of the DMM's 10 MΩ input resistance.

1. Series sources and KVL (Figure 4, R = 300 Ω)

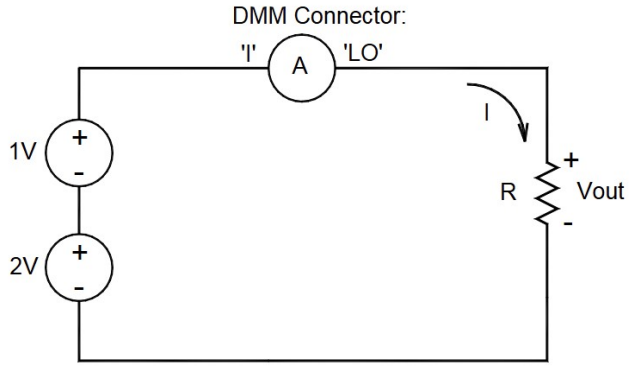


Figure 1: Two supply channels (1 V and 2 V) in series with R, DMM measuring current

> Q1: KVL around the loop: +1 V + 2 V − V_{out} = 0 ⟶ $V_{out} = V_1 + V_2$ = 3 V
> Expected current: I = 3 V / 300 Ω = 10 mA
> Q2: measured V_{out} = 2.99997 V; error = (3 − 2.99997)/3 × 100% = 0.001 %

Q3: The reading was positive (+2.99997 V), so the DMM's + lead was at the top of R and the − lead at the bottom, the same as the V_{out} marking in Figure 1.

> Q4: ±5% resistor: R = 285 Ω to 315 Ω
> I_{max} = 3 / 285 = 10.53 mA I_{min} = 3 / 315 = 9.52 mA
> DMM measured: I = 10.047 mA ⟶ inside the range

10.047 mA corresponds to R ≈ 298.6 Ω, which is very close to the nominal 300 Ω. My notes also show an earlier reading of 21 mADC. That doesn't fit 3 V across 300 Ω, so I used 10.047 mA.

2. Resistance measurements

Measurement	Value
R1 (step 15) / R2 (step 16)	98.86 Ω / 197.04 Ω
Series, measured (step 17)	295.71 Ω
Parallel, measured	65.93 Ω
1 Ω resistor with test leads (step 22)	1.19 Ω
1 Ω resistor, leads zeroed out (step 23)	0.973 Ω

> Q5: R_{series} = 98.86 + 197.04 = 295.90 Ω; difference = |295.90 − 295.71| / 295.90 = 0.06 %
> Q6: $R_{parallel}$ = 98.86 · 197.04 / (98.86 + 197.04) = 65.83 Ω (measured 65.93 Ω, 0.15 %)
> Q9: step 22: |1.19 − 1| / 1 × 100% = 19 %
> Q10: step 23: |0.973 − 1| / 1 × 100% = 2.7 %
> Q11: both leads = 1.19 − 0.973 = 0.217 Ω ⟶ each lead ≈ 0.109 Ω

Q7: Body resistance (step 19). My notes record 0.6 without a unit. Taking it as 0.6 MΩ (typical for dry skin): V = I·R = 0.010 A × 600,000 Ω = **6,000 V**. In reality skin breaks down at a much lower voltage, which is why we never touch anything above 42 V.

Q8: The two body-resistance readings (steps 20 and 21) were different and kept changing. Skin resistance depends on moisture, pressure and contact area. Wet skin and a firmer grip give lower resistance.

3. Powered circuits and the DMM's input resistance

Q12: No, the readings in steps 24 and 25 did not look correct. The DMM measures resistance by sending its own small test current. An outside source in the circuit adds to that current, so the reading is wrong, and the meter can be damaged. Resistance must only be measured with the power off.

> Q13: KVL: $V_S = V_{R1} + V_{R2} \longrightarrow 1.63\ V + 1.68\ V = 3.31\ V = V_S$

The measured voltages add up to the source voltage, so KVL holds. When the DMM measures across a resistor, its 10 MΩ input resistance is connected in parallel with it. That pulls a little current and lowers the reading. Since my readings add up exactly, R1 and R2 must be much smaller than 10 MΩ, so the effect is too small to see. With resistors in the megaohm range, the readings would come up short and seem not to add up.

4. Voltage divider

> $V_{out} = V_S \cdot R_2 / (R_1 + R_2)$

Q14	V_S	R_1	R_2	Target V_{out}	Measured V_{out}
My divider					

My notes do not have the values for this part yet.